Lec 2. Sampling distributions of $\bar{X}_n$ and S_n^2

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Learning objectives

After this section, you should be able to ...

- understand the distributions of $\bar{X}_n$ and S_n^2
- understand the delta theorem
- understand the orthogonal transformations of multivariate Normal vectors

1.1 Preliminary: χ^2, t and F distributions

Chi-square distribution

Definition

If $X_1, X_2, \cdots, X_n$ i.i.d. $\sim N(0,1)$, then $X = \sum_{i=1}^{n} X_i^2$ is distributed according to the chi-square distribution with n degrees of freedom. This is usually denoted as $X \sim \chi_n^2$.

The probability density function (pdf) of χ_n^2 is

$$g_n(x) = \begin{cases} \frac{1}{2^{\frac{n}{2}}\Gamma(\frac{n}{2})} x^{\frac{n}{2}-1} e^{-\frac{x}{2}}, & x > 0, \\ 0, & x \leq 0. \end{cases} \tag{1.1}$$

Chi-square distribution

Properties

Student t distribution

Definition

If $X \sim N(0,1), Y \sim \chi^2_n$, and X is independent of Y, then

$$T = \frac{X}{\sqrt{Y/n}}$$

is distributed as the *student* t distribution with n degrees of freedom. This is usually denoted as $T \sim t_n$.

The probability density function (pdf) of t_n is

$$t_n(x) = \frac{\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n}{2})\sqrt{n\pi}} \left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}}, \quad -\infty < x < \infty \tag{1.2}$$

t distribution

Properties

F distribution

Definition

If $X \sim \chi^2_m$, $Y \sim \chi^2_n$, and X is independent of Y, then

$$F = \frac{X/m}{Y/n}$$

has F distribution with n and m degrees of freedom. This is usually denoted as $F \sim F_{m,n}$.

If a random variable $F \sim F_{m,n}$, then the pdf of F is given by

$$f_{m,n}(x) = \begin{cases} \frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{n}{2})\Gamma(\frac{m}{2})} m^{\frac{m}{2}} n^{\frac{n}{2}} x^{\frac{m}{2}-1} (n+mx)^{-\frac{m+n}{2}}, & x > 0, \\ 0, & x \leq 0. \end{cases} \tag{1.3}$$

F distribution

Properties

1.2 Normal population case

Theorem 1. *If $X_1, \cdots, X_n$ are independent random variables with $X_k \sim N(a_k, \sigma_k^2)$, for $k = 1, \ldots, n$, and $c_1, c_2, \cdots, c_n$ are constants not all zero, then*

$$T = \sum_{k=1}^{n} c_k X_k \ \sim \ N\left(\sum_{k=1}^{n} c_k a_k, \sum_{k=1}^{n} c_k^2 \sigma_k^2\right)$$

In particular, when $c_1 = \cdots = c_n = 1/n, a_1 = \ldots = a_n, \sigma_1^2 = \ldots = \sigma_n^2$, that is, $T = \frac{1}{n} \sum_{i=1}^{n} X_i = \bar{X}$, we have

$$\bar{X} \sim N(a, \sigma^2/n).$$

Proof. It can be proved easily by the reproducibility of normal distribution, or using the characteristic functions. □

Theorem 2. *If* $X_1, X_2, \cdots, X_n$ *i.i.d.* $\sim N(a, \sigma^2), \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ *and* $S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ *are sample mean and sample variance, respectively, then*

(1) $\bar{X} \sim N(a, \frac{1}{n}\sigma^2)$;
(2) $(n-1)S_n^2/\sigma^2 \sim \chi^2_{n-1}$;
(3) $\bar{X}$ *and* S_n^2 *are independent.*
(4) $T_n = \frac{\sqrt{n}(\bar{X}-a)}{S_n} \sim t_{n-1}$.

Hints: orthogonal transformations of independent Normal variables.

Example

Let $X_1, \cdots, X_n$ $i.i.d \sim N(a, \sigma^2)$, and $\bar{X} = \dfrac{1}{n}\sum\limits_{i=1}^{n} X_i$, $S_n^2 = \frac{1}{n}\sum\limits_{i=1}^{n}(X_i - \bar{X})^2$, find the distribution of $\xi = (X_1 - \bar{X})/S_n$.

1.3 Asymptotic results for non-normal case

Theorem 3. *(Central Limit Theorem, CLT). Let $X_1, \ldots, X_n$ be an iid sample from population with mean μ and variance $\sigma^2 < \infty$. For $\bar{X}_n$ the sample mean, then as $n \to \infty$,*

$$\bar{X}_n \xrightarrow{a.s., P} \mu$$

$$Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{\mathcal{L}} N(0,1)$$

CLT

Theorem 4. *Let $X_1, \ldots, X_n$ be an iid sample from distribution with mean μ and variance $\sigma^2 < \infty$. For S_n^2 the sample variance, we have*

(1) $S_n^2 \xrightarrow{P} \sigma^2$

(2) $\sqrt{n}(S_n^2 - \sigma^2) \xrightarrow{\mathcal{L}} N(0, \mu_4 - \sigma^4)$, *where* $\mu_4 = E(X_1 - \mu)^4$.

Theorem 5. *Let $X_1, \ldots, X_n$ be an iid sample from distribution with mean μ and variance $\sigma^2 < \infty$. For $\bar{X}_n$ the sample mean and S_n^2 the sample variance, let $T_n = \frac{\sqrt{n}(\bar{X}-\mu)}{S_n}$, then $T_n \xrightarrow{\mathcal{L}} N(0,1)$.*

Theorem 6. *(Delta Theorem). For a sequence* $\{T_n\}$*, suppose there are numbers* θ *and* $v_\theta > 0$ *such that* $\sqrt{n}\,(T_n - \theta)$ *converges in distribution to* $\mathrm{N}\,(0, v_\theta)$*. Let* $g(t)$ *be a function, differentiable at* $t = \theta$*, with* $g'(\theta) \neq 0$*. Then* $\sqrt{n}\,[g\,(T_n) - g(\theta)]$ *converges in distribution to* $\mathrm{N}\left(0, [g'(\theta)]^2\, v_\theta\right)$*, as* $n \to \infty$*.*

Example

Let $X_1, \ldots, X_n$ be an independent sample from $\text{Exp}(\theta)$ whose mean is θ. For sample mean $\bar{X}$, find the asymptotic distribution of $\bar{X}^{1/2}$.